

Source-Aware Review of Concurrent Agent Outputs: A Computational Study of Familiarity, Capacity and Error

OverseeingManyLLMs research project

Separate computational draft; no participant observations

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Abstract

Concurrent language-model task workers can produce more material than a reviewer can inspect. Arranging questions around their shared source may reduce repeated orientation, but the same saving may be available through ordinary source-aware queue navigation. We study this distinction with a discrete-event simulation using 36 original TAT-QA questions, six real financial-report contexts and genuine saved model drafts. Reviewer behavior and arrivals are explicitly modeled rather than observed. A committed sensitivity design contains 66 settings and 31,680 paired runs. Grouping improves mean correct releases over arrival-order review in 24 settings, ties in two and reduces them in 40. It never exceeds a queue with the same bounded source-aware navigation and no grouping overhead. Costly orientation and short familiarity can favor reordering, whereas persistent familiarity, carryover errors and cheap manual construction weaken or reverse apparent benefits. The contribution is a reproducible conditional operating map and an information-equivalent control. It provides no evidence of measured human efficiency or workload improvement.

1 Question and evidence boundary

A review desk can preserve a source, an answer and its approval controls while other agent requests arrive. It can also group related questions. The research question is whether grouping improves useful completion within a fixed review budget, and which mechanism would explain that improvement. Showing several questions together need not be beneficial if a conventional queue can already visit them consecutively. Conversely, grouping could affect comprehension or navigation choices in ways a timing simulation cannot establish.

We therefore compare explicit navigation strategies under a common source-memory model. We neither grant grouping a speed multiplier nor withhold source reuse from a queue. Separate outcomes count correct releases, incorrect releases and unfinished work. More releases alone are not better oversight. A manual condition also tests the assumption that reviewing these particular drafts is useful.

The inputs come from the existing matched review study, whose prospective human protocol and position paper remain unchanged. TAT-QA combines financial tables and accompanying text with human-written questions and annotations[1]. We retain all 36 selected questions and actual replica-0 Qwen2.5-7B-Instruct drafts, including errors and missing answers. Only 7/36 drafts are jointly correct in answer and scale; five displayed fields are empty or unsupported. These values describe saved model outputs, not reviewer competence.

Evidence layer	Status in this study
Financial tables and text	Real report material represented by the benchmark
Questions and references	Original benchmark questions and annotation labels
Initial drafts	Actual saved generations; no new inference or artificial errors
Workers, arrivals and cutoff	Constructed supervision application
Review timing and outcomes	Simulated under analyst-selected assumptions
Human behavior	Unmeasured; no participants, ratings or practice records used

2 Shared reviewer and event protocol

A question is one user-valued goal. All requests remain accounted for from offer through release or cutoff. Receipt during a review cannot replace the active artifact. Approval and release are distinct actions bound to an exact output version. A correction creates a new version. Deferral retains the original output; rejection and approval without release remain unfinished. The simulator calls the existing review engine’s transition and invariant functions. It does not implement a looser simulation-only release rule.

Review time is the sum of orientation, question interpretation, verification or construction, decision interaction and release interaction. A grouped session adds its declared overhead. For source s , orientation is Sf_s on a first visit. On return it is

$$O_s = Sf_s \left[r + (1 - r) \left(1 - e^{-g_s/\tau - k_s/\kappa} \right) \right], \quad (1)$$

where g_s is elapsed simulated time since the last completed review decision on that source, k_s counts intervening decisions, and $r = 0.1$. The public source-size factor f_s is character count divided by 3,200, clipped to $[.75, 1.25]$. All six selected sources have $f_s = .75$, so their first-visit orientation is $.75S$; the size factor does not discriminate difficulty in these matched packets. There is no additional switching charge. Consecutive FIFO questions receive the same familiarity saving as grouped questions. A completed failed construction can also refresh source familiarity.

Question, verification and construction times use a common wording factor $1 + \min(40, \text{word count})/80$ and mean-one lognormal jitter with $\sigma = .2$. Neither the timing rule nor the navigation policy receives gold derivations or future outcomes. Random draws are keyed by seed, question, attempt and component, so matched methods share potential outcomes wherever their paths permit.

Assumed component	Reference and declared sensitivities
Orientation coefficient S	25 s; also 0, 5, 75 s
Memory τ, κ	600 s, 8 decisions; also 45 s, 2; effectively persistent
Question / verification	8 / 18 s; verification also 8, 40 s
Detected-error construction	45 s
Manual or empty-draft construction	45 s; also 20, 90 s
Decision / release interaction	2 / 2 s
Grouped-session overhead	4 s; also 0, 12 s
Deferral	Eligible again after 45 s, at most two attempts

2.1 Fallible outcomes and limited authority

The simulated environment uses annotations to determine whether the initial draft is correct. A wrong nonempty draft is detected with probability d and successfully corrected with probability c after detection. Undetected errors are approved incorrectly. A correct draft can be damaged with probability h . Manual and empty-draft constructions succeed with probability m . Successful simulated constructions copy only that question’s annotated answer and scale. An unsuccessful construction uses an explicit simulation sentinel, not a generated financial assertion. Half of failed constructions recognize inability and defer, then reject on a second recognized failure. Other failures release a wrong answer.

The ideal setting is $(d, c, h, m) = (1, 1, 0, 1)$. Higher and limited effectiveness use $(.85, .85, .02, .75)$ and $(.55, .60, .08, .50)$. A separate stress hazard of 0, .08 or .20 can spoil an otherwise correct answer after another question from the same source. It applies equally to all methods. These are assumptions, not inferences from model accuracy or fitted measurements of financial reviewers. Source grouping never grants shared approval authority.

3 Comparisons, design and numerical uncertainty

Policy	Public next-request rule
FIFO Q	Earliest received eligible request, then stable question identifier
Grouped G	FIFO starts up to three available questions from one source; finish that set, then return to FIFO. Charge overhead for a multi-question group.
Source-aware Q	G’s identical bounded selection, without grouping cards or overhead
Sticky Q	Continue on the last source while eligible related questions remain
Manual M	FIFO selection and source-only construction, with no draft

All selectors see arrived identifiers, actual source identity, receipt times and pending status. They cannot see correctness, future arrivals or random outcomes. No policy waits to assemble a group. G assumes use of grouping whenever possible; it does not model observed uptake. FIFO describes a strategy, not a constraint of the real queue interface, where a person can already select related questions.

The original scenario offers twelve questions for 540 seconds. Arrivals reproduce the matched schedule, at 0, 4, . . . , 20 and 180, 184, . . . , 200 seconds plus .25 s delivery. Each of three packets has two sources with six questions each. Separate stress scenarios use 6, 12, 24 or 36 distinct questions, 270, 540 or 900 s capacity, burst/two-wave/spread arrivals and interleaved/source-blocked order. Smaller loads use declared public prefixes; larger ones union packets without duplicating goals. Costs, effectiveness, memory and carryover are varied in prespecified compact subsets, not a fitted distribution of real workloads.

The frozen design comprises 66 settings, three rotations, 32 seeds and five policies, totaling 31,680 runs. A preliminary 64-seed forecast was reduced before comparative outcomes; all parameter settings were retained. The sources were previously inspected and do not establish six independent reports. Rotations, questions and seeds are not people or new organizations.

For each setting, paired rotation differences are averaged within seed. Intervals are the mean plus or minus 1.96 Monte Carlo standard errors across 32 seed blocks. They quantify numerical integration uncertainty conditional on the fixed model and sources. More seeds cannot validate a memory function or reduce uncertainty about an unmeasured reviewer population. We report each setting separately and do not pool the sensitivity grid into a global effectiveness claim.

4 The original setting favors simpler controls

At $S = 25$ s (18.75 s on first visit), long memory and higher effectiveness, G minus Q is -0.156 correct releases per twelve offered, with Monte Carlo interval [-0.270, -0.043]. G minus source-aware Q is -0.115 [-0.170, -0.059]. Seed-block G–Q wins/ties/losses are 3/16/13.

Policy	Correct releases	Incorrect releases	Unfinished
FIFO Q	5.45	1.06	5.49
Grouped G	5.29	1.04	5.67
Source-aware Q	5.41	1.06	5.53
Sticky Q	5.61	1.03	5.35
Manual M	5.18	0.79	6.03

Every offered question remains in this table. An answer must be jointly correct in content and scale, approved and released at its current version. Construction failures, never-selected requests, rejections and cutoff work are retained. The manual comparison is small and numerically uncertain here: Q–M is +.271 [–.007, .548] and G–M is +.115 [–.152, .381] correct releases. These intervals are not evidence of human equivalence or of preserved quality. Q–M holds FIFO navigation fixed. G–M additionally changes ordering, so it compares whole workflows rather than isolating draft availability.

5 Conditional operating regions

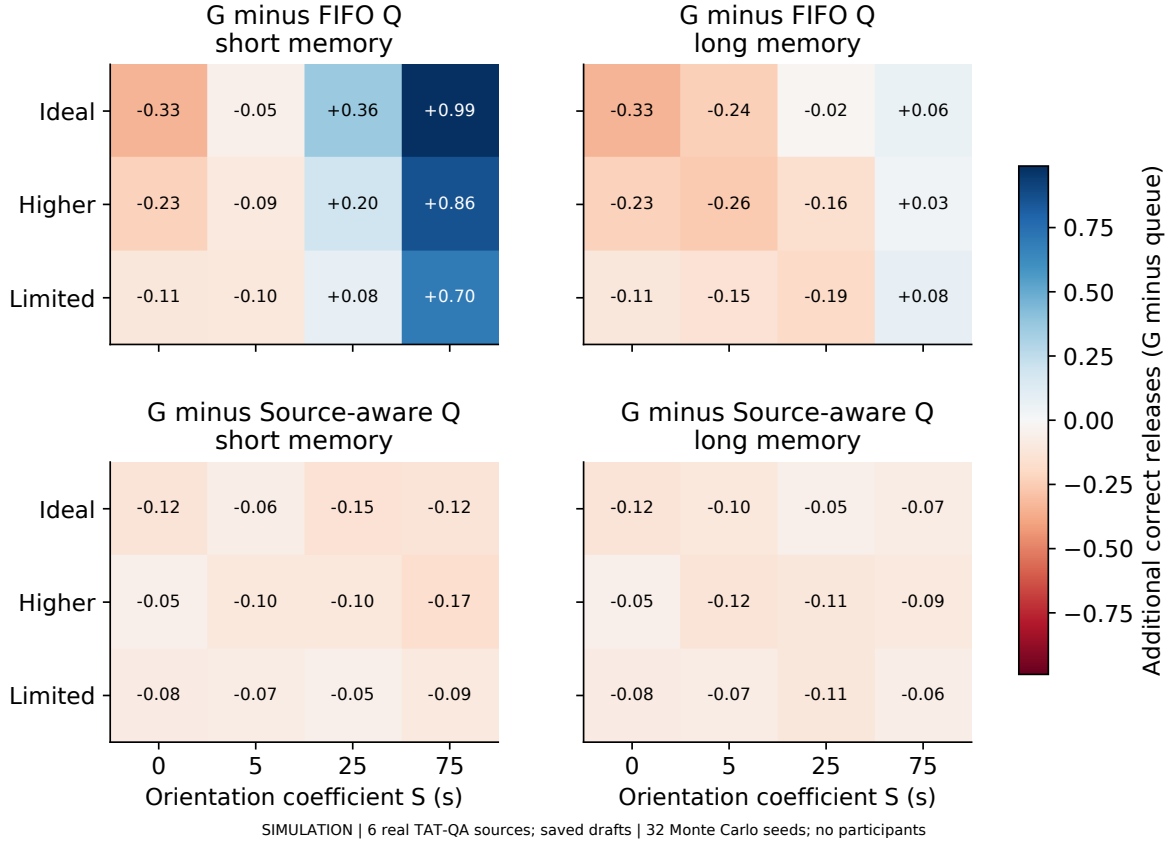


Figure 1: Simulated additional correct releases for G under the original workload. Real TAT-QA sources and actual drafts are fixed inputs; every timing and reviewer parameter is assumed. Blue favors G and red favors the indicated queue. Both rows use the same centered scale. Means average three packets and 32 Monte Carlo seeds, not human participants.

With $S = 25$ s, shortening memory changes G–Q from -0.156 to $+0.198$ [$+0.081$, $+0.315$]. At $S = 75$ s with short memory it is $+0.865$ [$+0.727$, $+1.002$]. That gain also increases wrong releases from .688 to .833. More correct releases can therefore accompany greater risk. Ideal-reviewer cells expose the capacity effect without error uncertainty, while fallible cells show whether it persists under the assumed detection and repair rates.

Across the declared grid G has positive, tied and negative mean differences against FIFO in 24, 2 and 40 settings. Against bounded source-aware Q the counts are 0, 4 and 62. Three ties are the zero-overhead cases, where equality follows mechanically from shared selection and random inputs. The fourth has six questions and a fifteen-minute budget. All assisted policies then average 5.031 correct, .938 incorrect and .031 unfinished. This is ample capacity with shared fallibility, not proof of equivalent human interfaces.

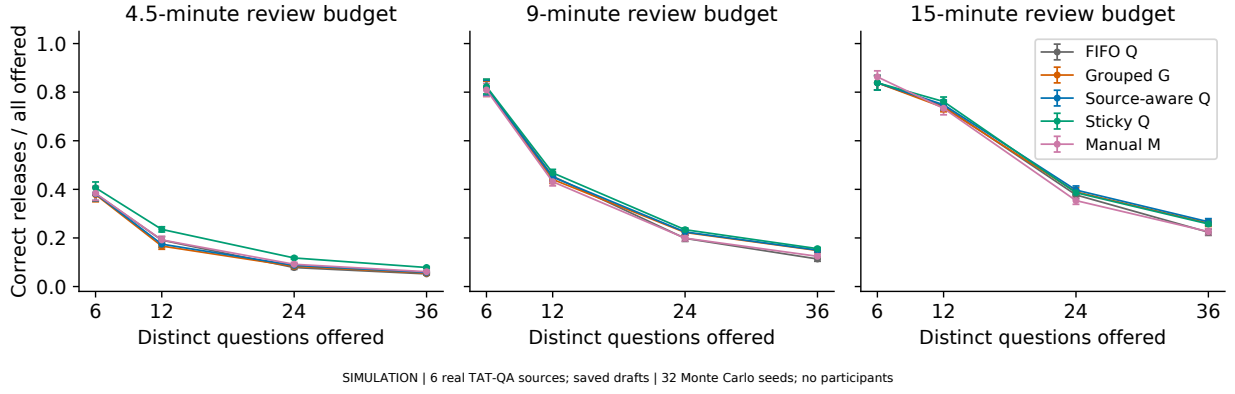


Figure 2: Simulated correct releases divided by all distinct offered questions. Arrivals, capacity and reviewer behavior are constructed. Error bars are conditional Monte Carlo intervals across 32 seeds, with three source-packet rotations per seed. No point represents a participant. Lines connect the declared load settings.

Higher demand creates more opportunity for ordering to change the completed set. At 36 questions and fifteen minutes, G-Q is $+1.333$ $[1.068, 1.599]$ answers, but G still trails source-aware Q by .208. With 24 questions in two waves, interleaving yields $+.583$ $[\cdot462, \cdot704]$ for G-Q, whereas source-blocked ordering yields $-.042$ $[-.090, .007]$. A FIFO queue already adjacent by source leaves little repeated orientation to save.

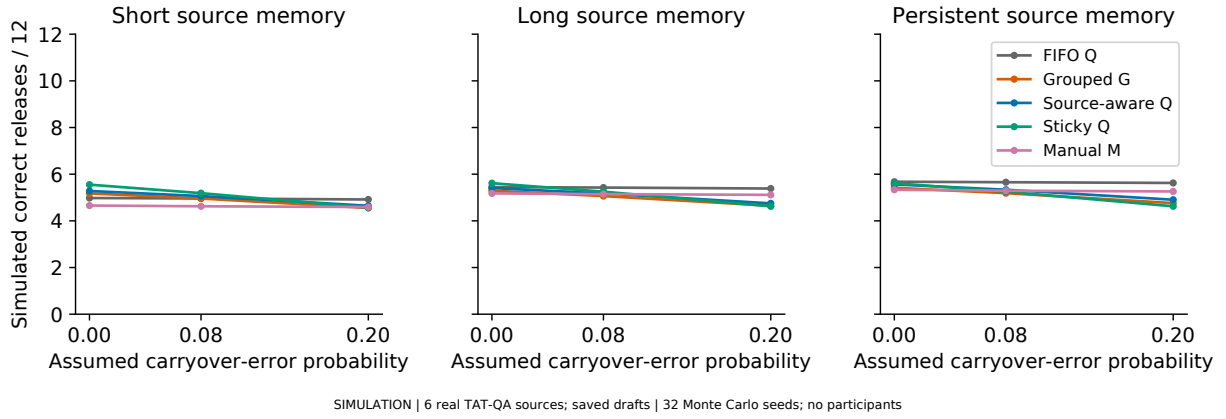
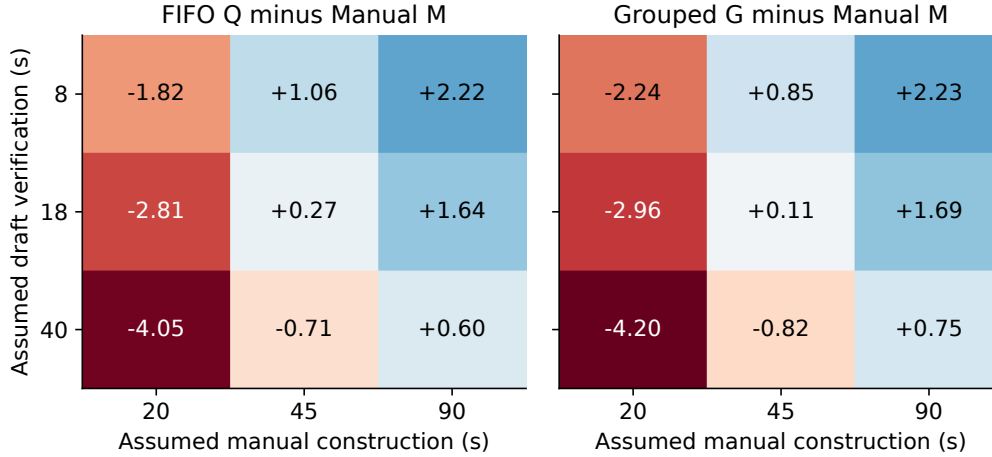


Figure 3: Simulated correct releases under different memory and same-source carryover assumptions. The hazard applies to every method, not just grouping. Real source material does not validate these behavioral probabilities.

Carryover challenges the same mechanism that makes consecutive-source work efficient. With long memory and hazard .20, G-Q becomes $-.740$ $[-.925, -.555]$. Wrong releases rise from 1.042 to 1.688 for G, compared with 1.062 to 1.125 for FIFO. Sticky-source review can also lose its advantage. This is a model-based exposure effect, not evidence that financial reviewers actually make these carryover errors.

5.1 A conditional break-even explanation

For the same attempted work and success path, saved review time is orientation saved minus grouping overhead. That balance matters only if it permits another full decision and release before cutoff. Summing orientation over different completed subsets is not a valid causal explanation by itself. Delayed unrelated work and harmful carryover also matter. At zero overhead, G and source-aware Q are identical by induction over event times. With positive overhead, changing eligible sets can change paths, so we assert no general dominance theorem. The finite-grid result supports navigation as the simpler explanation, without proving that real cards have no comprehension benefit.



SIMULATION | 6 real TAT-QA sources; saved drafts | 32 Monte Carlo seeds; no participants

Figure 4: Simulated assistance effect in correct releases relative to manual work. Positive values favor drafts; negative values favor manual construction. Inputs are genuine imperfect drafts, while verification, construction and reviewer effectiveness are analyst assumptions. Both panels use one symmetric color scale.

6 Interpretation, limitations and next evidence

When manual construction is cheap and checking is expensive, source-only work outperforms reviewing these drafts. $G-M$ is -4.198 $[-4.528, -3.868]$ at 20-second manual construction and 40-second verification. At 90-second manual construction and 8-second verification it is $+2.229$ $[2.040, 2.418]$. Those declared extremes identify a boundary, not a human assistance effect.

The central contribution is a controlled conditional comparison that separates source-aware navigation from grouping overhead while retaining real questions, wrong drafts, unfinished work and exact-version releases. Queues, batching and memory models are established mechanisms. Interruption research distinguishes returning to a task from recovering context[2]; scheduling research already studies setup/batching tradeoffs[3]. Neither source calibrates this financial-review model. Our function, numerical ranges, feature-use rule and question-independent random errors remain unvalidated. Shared errors, skill differences, evidence understanding and adoption choices could alter the outcomes.

All 31,680 command streams replayed, 200,286 released answers were independently re-scored, and 120 traces matched the ordinary atomic review engine. Twelve focused tests check information, timing and authority boundaries. The 52 protected human study and manuscript files are unchanged. These checks establish software accounting, not reviewer comprehension. The CPU batch took 265 seconds with no new inference. Saved commands, inputs, configurations, figures and reproduction instructions accompany this separate draft.

The next observations should measure orientation after consecutive and intervening questions, actual grouping interaction, source-aware queue navigation, detection and repair, manual construction and unit carryover. They can calibrate or falsify the modeled operating regions. The preserved prospective study remains useful; any added automatic source-aware comparator needs a separately declared design. This computational study supports a systems-and-simulation finding, while human correctness, efficiency, control and workload benefits remain unmeasured.

References

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